

# Edouard Murat

## Gameplay Programmer & Team Lead

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French: fluent • English: fluent

### PROFILE

Gameplay programmer and team lead with 20 years of software experience, including five years of production C#. Led a six-person Godot team building curriculum-aligned educational games for Android. Specializes in gameplay systems, AI behavior, procedural generation, shader programming, and reusable game frameworks. Shipped games on Steam, Google Play, and the Apple Store.

### TECHNICAL SKILLS

|                    |                                                                                                                         |
|--------------------|-------------------------------------------------------------------------------------------------------------------------|
| Game Engines       | Godot, Unity                                                                                                            |
| Languages          | GScript, C#, Python, Java, HTML/CSS/JS, SQL                                                                             |
| Shaders & Graphics | Shader scripting (Godot, Unity)                                                                                         |
| Gameplay Systems   | Input (keyboard & mobile), AI behaviors, procedural generation, physics-based mechanics, state machines, combat systems |
| Tools & Frameworks | Git, GitHub, Blender, VS Code, Django, SQLite, Audacity, Linux                                                          |
| Other              | Agile workflows, UI integration, performance optimization                                                               |

### EXPERIENCE

#### Lead Gameplay Programmer (Godot) — Sigma Digital, South Africa (remote) Aug 2025 – Aug 2026

Led a six-person Godot team building curriculum-aligned literacy games delivered on Android tablets to first- and second-grade children in Africa.

- Hired the programming team and led it as technical lead, owning architecture, code review, and sprint planning
- Designed the platform's UI framework and an MVVM + FSM architecture adopted across all activities, decoupling game logic from presentation, enabling per-activity unit testing, and making curriculum content swappable without rebuilding core mechanics
- Delivered six mini-games and supported two rounds of playtesting with the target age group
- Built for low-cost Android tablets, with gameplay mapped to a first- and second-grade reading and writing curriculum
- Worked directly with educators and instructional designers to translate curriculum goals into game mechanics, bridging educational theory and interactive design across each sprint
- Provisioned GPU-enabled AWS EC2 instances to run automated Godot UI tests, enabling headless rendering for CI without local hardware

#### Gameplay Programmer (Godot) — Shattered Paradise, One More Turn Studios (remote) Jul 2025 – Dec 2025

Sole programmer on a turn-based grid-based dungeon crawler roguelike (Dante's Inferno theme, procedurally generated dungeons). The creative director defined the vision; owned all technical work end-to-end.

- Wrote all Godot gameplay code, shaders, the game website, and backend from scratch
- Prioritized and built features independently across the full stack

#### Gameplay Programmer (Godot) — Pursuit of Life, Dreamboat (remote) Jun 2025 – Aug 2025

Shader-focused contract on a pixel-art urban life-sim.

- Built a dynamic rain system with customizable impact zones and per-region wetness tracking
- Implemented real-time puddle reflections for sprites, NPCs, and swaying foliage
- Developed day/night lighting driven by a global time-of-day system

#### Game Developer — EdNoKa, Montreal 2022 – Present

Founder and lead developer of an educational gaming platform blending custom-built quizzes with gameplay across farming, hack & slash, platformer, puzzle, and sports genres.

- Published 5+ games as public Steam playtests using Unity and Godot, integrating real-time quizzes into diverse gameplay loops
- Designed and implemented gameplay systems: player control, AI, procedural generation, level logic, scoring, and feedback
- Implemented unified input systems supporting both keyboard and mobile touch controls
- Created custom shaders for visual storytelling and UX (X-ray vision, mini-cameras, 3D text, underwater effects)
- Built a full-stack Django backend: authentication, data storage, in-game unlockables, and content delivery
- Built and maintained deployment pipelines for Steam, Google Play, and the Apple Store; games live on the public Steam playtest

#### Web Developer and Graphic Designer — HeadPause, Montreal 2018 – 2020

- Developed and maintained the company website and online presence
- Created animations and visual assets, and ran social advertising campaigns

## Java and Web Developer — BMC Software, Montreal

2013 – 2014

- Partial management of the development team in SCRUM, including code reviews and TDD
- Built a new adaptive website to client requirements alongside parallel remote teams
- Interviewed candidates for web developer positions

## C# Developer and Analyst — Fujitsu Consulting, Montreal

2008 – 2013

- Five years of production C#: a payment solution for HydroSherbrooke and the daycare management system for the Ministère de la Famille du Québec
- Analyzed client needs, resolved blocking issues, and trained new developers across a three-year project

## SELECTED GAME PROJECTS

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*Full list of playable games at [edouardmurat.com](http://edouardmurat.com)*

### Base of 30 (Godot) — Rogue hack & slash through procedurally generated rooms

2025

- Designed and implemented all core gameplay systems, including combat and player progression
- Developed procedural generation for room layouts, obstacle placement, and navigation paths
- Built custom behavior trees for enemy types with unique attack patterns and movement logic

### Aerial Ambush (Godot) — Skill-based platformer with homing missiles

2025

- Implemented responsive, high-speed player movement with precise collision handling
- Developed homing missile AI with adjustable targeting and physics-based steering
- Designed and balanced dodge-based gameplay loops for short sessions and high replayability

### Agent Ka (Godot) — Waldo-like game in a generated city with crowds and powers

2024

- Built a procedural city generation system with randomized buildings, roads, and crowds
- Programmed crowd simulation with dynamic movement, behaviors, and density-based logic
- Implemented X-ray, mini-camera, and slow-motion powers via custom shaders

### Veggies Island (Godot) — Farming game on a cartoonish island

2024

- Implemented the core farming loop: planting, watering, harvesting, and crop growth stages
- Built procedural island generation with randomized terrain, resources, and farm zones
- Developed a lightweight inventory system for harvested crops and seeds

### Pico Golf (Godot) — Mini golf challenges with physics-based terrain

2023

- Designed 3D mini-golf courses with increasing difficulty and varied sea-floor layouts
- Integrated animated fish and custom water and caustics shaders to enhance immersion
- Developed audio-visual feedback systems for a calming, low-pressure experience

### Tomb (Godot) — Doom-inspired shooter for Godot Wild Jam #42

2022

- Built from scratch in under one week during a jam, demonstrating rapid prototyping
- Engineered arcade-style shooter feel: player movement, shooting, and health systems
- Designed enemy AI using state machines and pathfinding logic with trap systems

Earlier titles include Left 4 Dead 2D, Meditation Card Game, Fruit, Archer, and Meditation World VR.

## EDUCATION

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### B.A. Psychology — Concordia University, Montreal

2015 – 2017

### B.Eng. Computer Engineering — University of Sherbrooke

2005 – 2009

### College Diploma, DEC in Sciences — Stanislas College, Montreal

2003 – 2005

## CERTIFICATIONS

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### Certified Associate Python Programmer — Python Institute

2024

### Oracle Professional Certification, Java SE 7 Programmer

2013

### Oracle Associate Certification, Java SE 7 Programmer

2012

References available upon request.